

THE 30 DOLLAR STOVE FAN THAT NEEDS NO PLUG

Your wood stove heats the ceiling while your feet freeze. This fan pushes that heat sideways, powered by the stove itself, even in a blackout.

A heat powered stove fan sits on the stove top, and the stove's own heat spins it, with no battery and no plug. Four blade versions sell for about 30 to 40 dollars. In one full winter test, a fan like this warmed the far corner of the room by about 6 degrees.

WHY YOUR FEET STAY COLD

72°F	Thermostat setting, 48 inches up the wall, in a typical winter room described by building science writer Martin Holladay.
65°F	The air down by your feet.
79°F	The air up near the ceiling. A 14 degree spread in one room, at the same moment.
About 5%	How much lighter air gets warmed from 70 to 100°F. Air off a 500 degree stove top is far lighter and goes straight up.
About 5.5°	The spread between head and ankles the building comfort standard allows. Past that, people report cold feet.

HOTTER ISN'T THE ANSWER

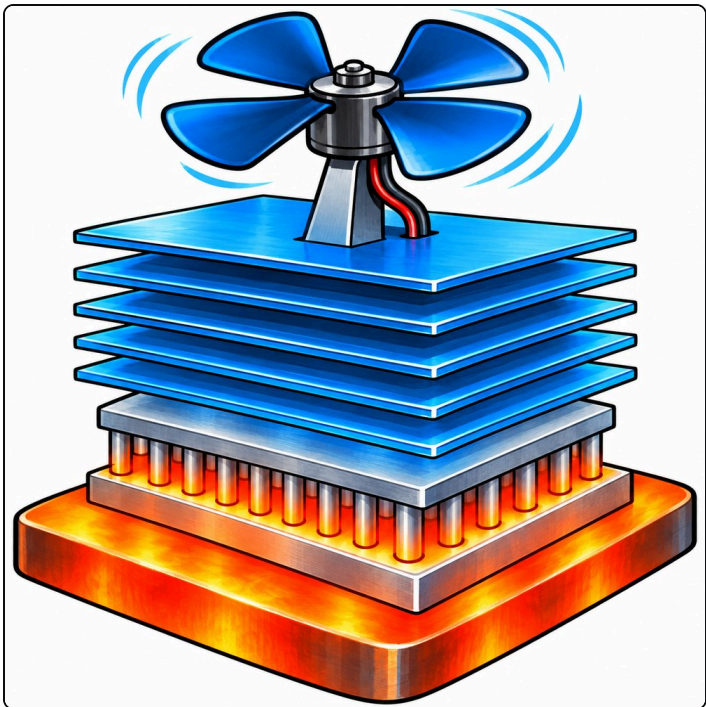
- ☐ Good stove top zone: about **400 to 650°F** on a magnetic stove top thermometer.
- ☐ Below 300°F the fire smolders and coats the chimney with creosote.
- ☐ Above 700°F the stove is over firing. Sustained over firing can warp the baffles and the steel, and most warranties won't cover it.
- ☐ A cord of seasoned hardwood runs about 300 dollars on average, more in the Northeast. A home heating mostly with wood burns 3 to 6 cords a season: 900 to 1,800 dollars, and a fair share of it heats a ceiling.



The fan on a back corner of the stove top, off to one side of the pipe, thermometer beside its base.

SHOPPING LIST

- ☐ **Stove fan:** about 30 to 40 dollars, four blades, with a **bimetal safety strip**. Skip any fan with no overheat protection.
- ☐ **Magnetic stove top thermometer:** about 10 to 15 dollars. Don't run the fan without one.
- ☐ **Two cheap indoor thermometers** for the test on page 3.
- ☐ **A stove glove** for moving the fan.
- ☐ **Big room?** A second fan. Two 30 dollar fans still cost less than the cheapest plug in blower kit.



Hot base below, cool fins above. The module runs only on the gap between them.

HOW IT RUNS WITH NO PLUG

- ☐ Heat one side of a joint between two different materials and you get a small voltage: the **Seebeck effect**. It's how a gas water heater's pilot sensor knows the flame is lit.
- ☐ Under the base: a flat square module of **bismuth telluride**, about an inch and a half across, packed with over a hundred pairs of tiny pillars wired end to end.
- ☐ Hot bottom, cool top: it makes a volt or two, enough to spin a tiny motor and aluminum blades.
- ☐ **It runs on the gap, and only the gap.** Both faces at the same temperature means zero output, however hot.
- ☐ The aluminum base soaks up stove heat. The fins shed heat. The blades sit right in front of the fins, so the fan's own breeze keeps its cold side cold. Hotter stove, faster fan, all on its own.

120 to 230°F	Stove top temperature where most fans start turning. Your model's start temperature is printed on the box.
400 to 480°F	Where they reach full speed.
About 650°F	Typical rating at the base. Some fans cut out a little lower.
About 645°F	Where a bimetal strip bends and tips the fan up on one edge. The base can still run too hot, so keep watching.

WHAT IT REALLY MOVES

- ☐ Listings claim about **100 to 240** cubic feet per minute. Bob Vila's test team ran four fans on a real stove for three weeks and wrote that the maker numbers aren't always accurate.
- ☐ A good fan on a hot stove blows out a match at about 2 feet. Past that, you won't feel a breeze. **It's a circulator, not a blower.**

Heat per hour (BTU) = airflow (cfm) × degrees warmer than the room × 1.08

Say a fan really moves 100 cfm of air 20 to 30 degrees warmer than the room: roughly **600 to 950 watts** pushed sideways instead of up. A generous upper bound, not a promise. The fan makes none of that heat. The stove already made it.

THE FULL WINTER TEST

344 sq ft	Room with a cast iron stove. Thermometer in the far corner, about 13 feet away.
66 to 72°F	Far corner, fan off versus fan on.
About 2° cooler	Right next to the stove, because the heat left faster.
14 to 5	Degree spread across the room, before and after.

SAVINGS, HONESTLY

- ☐ The 40 percent wood savings in ads? The same retailer calls it nonsense. Realistic: **5 to 15 percent** over a season. Caframo, maker of the Ecofan, claims up to 18 percent from a test it paid for. Treat that as a best case.
- ☐ Burn 4 cords at 300 dollars: 5 percent saves 60 dollars, one season pays for the fan. 15 percent is 180 dollars a year.
- ☐ **Savings only show up if you burn less.** Once your chair is warm, turn the air control down a notch, or put one less split on the next load.
- ☐ Caframo rates its module for over 10 years and its motor for 3 to 5 burning seasons.

CAN YOUR STOVE RUN ONE?

- ☐ Needs a flat, bare top that gets hot.
- ☐ Fireplace inserts with no open top won't work. Most pellet stoves don't get hot enough on top.
- ☐ Soapstone tops run cooler, so the fan may turn slowly. Curved or ribbed tops won't give the base full contact.

PLACEMENT

- 1** Flat spot at a **back corner** of the stove top, off to one side of the pipe.
- 2** At least a **hand's width** from the stovepipe. The pipe's heat cooks the fins.
- 3** Not dead center, usually the hottest spot on the stove.
- 4** Open space behind it, so it pulls cool air across the fins.
- 5** Point the blades toward where you sit, or at the doorway to the cold rooms, and leave those doors open.
- 6** Stick the thermometer on the stove top right beside the fan base.

THE TWO THERMOMETER TEST

- 1** Tape one thermometer at ankle height in the far corner, one high on the wall near the ceiling.
- 2** Alternate evenings: fan off, fan on, off, on.
- 3** Same size loads, same air setting each night.
- 4** Write down both readings every hour.
- 5** If the gap shrinks by **5 degrees or more**, the fan's doing its job.

SAFETY RULES. NOT OPTIONAL.

- ☐ Top climbs past **650°F**? Close the air control first and follow your stove manual. That's the real fix. The safety strip only lifts one edge a little.
- ☐ Past the fan's rated limit, move it to a cooler spot or off the stove. Lift it the way the maker says, usually by the handle, **never by the hot base**, wearing a stove glove.
- ☐ Set it down on a bare stone hearth, **never on a wood floor or a rug**.
- ☐ Keep hands, kids, and pets away from the blades.
- ☐ Don't drop it. The module is a ceramic sandwich, and a fall onto brick can crack it.

KEEP IT RUNNING

- ☐ Blow the dust off the fins at the start of each winter. Dusty fins stay hot and slow the fan.
- ☐ If it starts needing a nudge to get going, the motor's wearing out.
- ☐ One seller says cheap fans without a bimetal strip tend to die within a season or two.



Ice storm, grid down, and the one fan on the street still turning.

WHEN THE POWER DIES

An ice storm takes out the grid, and every furnace on the street goes quiet. You've got a wood stove, so you've got heat. But the usual ways to spread that heat need an outlet.

- ☐ **Blower kit:** 100 to 375 dollars, plugs into the wall, and only fits stoves built with a jacket for it. Stops in a blackout.
- ☐ **Ceiling fan on reverse:** helps on a normal night for maybe 50 watts, pulling the warm layer down. Needs power too.
- ☐ **Stove top fan:** turns as long as the top stays over its start temperature. It evens out the stove's room and helps nearby rooms a little. It won't heat a far bedroom by itself.

Run both on a normal night. Only one of them keeps going in a blackout.

NEXT STEP

At 3 in the morning the coals fade, the stove top drops below the start temperature, and the blades stop. Holding heat until morning takes something that stores it: watch the 300 dollar sand drum video next.

Watch the full video: youtube.com/@SketchySurvival101

More free sheets: sketchysurvival101.com/free

Subscribe so the next blackout does not catch you flat footed.